ACCRA-FLOW

A Modular Geospatial AI Framework for Paratransit Optimization

and Urban Resilience in the Greater Accra Metropolitan Area

Case Study Group 5: Urbanization and Infrastructure Resilience
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Executive Summary

Every day, Accra grinds to a halt. Not because the city lacks roads, but because it lacks information.

Thousands of TroTros and Okadas, the minibuses and motorcycle taxis that carry the majority of Greater Accra's commuters, operate in near-total informational blindness. Drivers choose routes by instinct. Intersections treat a 28-passenger minibus and a private saloon car as equivalent. Commuters board with no idea when they will arrive. The annual cost is approximately $301.5 million, roughly 8.21% of Ghana's GDP.

The long-term answer is clear: more roads, better infrastructure, sustained investment in formal transit. But that work is generational. Accra-Flow asks a narrower and more urgent question: given the network that exists today, how much of that loss is recoverable through better information alone? The answer, it turns out, is substantial.

The framework gives drivers real-time demand signals, gives intersections the intelligence to prioritise a loaded minibus over an empty car, and gives commuters a reliable journey estimate before they step onto the kerb. It does this through three layers: smart roadside cameras that process footage on-device to protect privacy; a self-learning signal system where intersections cooperate as autonomous agents to maximise passenger throughput; and a communication layer built for Accra's reality, USSD alerts that work on any handset, and mobile-money fuel rebates that reward driver cooperation directly.

This report documents the full journey behind Accra-Flow, from quantifying mobility quality in an informal transit context, through the failures that forced each design leap, to the architecture now ready for pilot deployment on the Madina-Tetteh Quarshie corridor.

2. The Problem Domain: An Economic and Human Emergency

2.1 The Scale of Congestion

Traffic congestion in Accra is not just an inconvenience: it is a structural drag on national development. The $301.5 million lost annually to gridlock disaggregates into three channels:

Cost Category	Annual Impact (USD)	Share of Total
Time lost in traffic	$214.3 million	71%
Stress-driven productivity decline	$54.6 million	18%
Fuel waste from idling vehicles	$29.1 million	11%

Cost Category	Annual Impact (USD)	Share of Total
Total annual loss	$301.5 million	100%

Table 1: Annual economic losses due to urban traffic congestion in Accra (GH₵ figures converted at ~GH₵14.93 per USD). Source: MyJoyOnline [5]; cross-referenced with World Bank Urban Mobility Report [32].

Beyond economics, the crisis is a public health emergency. Air pollution from aged vehicle engines idling in traffic cost Ghana an estimated $1.6 billion in 2019. In the Kaneshie district, fine particulate matter (PM2.5) levels reach ten times the WHO 24-hour safe limit [4]. Road-traffic injuries claim 10,000 lives annually and consume a further 1.6% of national GDP [7].

2.2 Who Bears the Burden

Aggregate figures obscure where the pain falls. Accra-Flow's design was anchored to four archetypes representing the modal experience of Greater Accra's commuter population:

Archetype	The Mechanism of Harm
Students (e.g., Abena, University of Ghana, Legon)	Loses approximately 10 hours per week to unpredictable journey times. Without real-time transit information, departure times are governed by guesswork, systematically eroding academic performance and well-being.
Perishable goods traders (e.g., Auntie Efua, Makola Market)	Reports up to 14.4% post-harvest loss due to transportation delays. Congestion translates directly into spoilage, higher food prices, and lost livelihoods, making this a food security crisis as much as a mobility one.
TroTro drivers (e.g., Kojo)	Burns a significant share of daily gross revenue in idling fuel. Operating on thin margins in ageing vehicles, a single mechanical failure can tip a driver into debt, yet these operators carry the vast majority of Greater Accra's commuters.
Private commuters (e.g., Mensah, Airport Residential)	Loses an estimated 15% of productive commute time to congestion that coordinated signal priority could eliminate. Though less economically vulnerable than other archetypes, private car users occupy disproportionate road space per passenger carried; reducing their journey times through network coordination is the most politically viable case for the framework's adoption.

Table 2: Socio-economic archetypes illustrating the differential burden of congestion.

These archetypes were not ornamental. They became the design constraints that every technical decision was tested against: any solution that did not work on a basic feature phone, or that transferred financial risk onto an already-marginal driver, would fail at the point of adoption.

3. Design Iterations

3.1 Iteration One: The Commuter Fluidity Index

The team's starting question was how to measure mobility quality in a system where a private car and a TroTro carrying 28 passengers share the same road. Conventional metrics, such as vehicles per hour and average speed, treat all vehicles as functionally identical. This is analytically wrong: a solution that optimises for vehicle flow rather than person flow will always disadvantage high-occupancy transit.

The response was the Commuter Fluidity Index (CFI): $CFI = \Sigma(P_i \times V_i) / (T_wait \times C_fuel)$. Each vehicle is weighted by its passenger count (P_i), giving a TroTro approximately fifteen times the mobility value of a private car. Velocity (V_i) captures productive movement; hub delay (T_wait) penalises time lost at stops; and the fuel idling coefficient (C_fuel) reflects the real money burnt by drivers waiting in queues.

The CFI was a genuine conceptual advance, but simulations exposed two fundamental limitations. First, the Balloon Effect: optimising a single intersection in isolation displaced bottlenecks rather than resolving them. Second, the Fixed-

Route Problem: even where the CFI identified a faster detour, drivers would not take it without real-time demand visibility. These two failures defined the brief for the next iteration.

3.2 Iteration Two: The TroTro Priority Algorithm

The TroTro Priority Algorithm (TPA) replaced static traffic signals with smart signals that understand which vehicles matter most. Using roadside cameras to estimate occupancy, the TPA allocates signal priority to the lane that can move the greatest number of people through the intersection. A dynamic Priority Score is computed in real time:

Component	Function
$n_v \times W_v$ (vehicle count × occupancy weight)	TroTros weighted at 15, private cars at 1.2, Okadas at 1.0, directly encoding person-throughput priority.
η_v (efficiency coefficient)	Ratio of current speed to optimal flow speed. Rewards moving lanes; penalises stalled ones.
$\delta(t_wait)$: starvation penalty	Increases a lane's priority score as its waiting time grows, preventing side streets from perpetual starvation.
$\varphi(v_drift)$: touting penalty	Subtracts priority if a vehicle is detected crawling below 5 km/h outside a designated hub, the signature of touting behaviour.

Table 3: Components of the TPA Priority Score.

Despite its advances, the TPA remained a system of human-designed rules. Its parameters could not adapt autonomously to conditions outside their design envelope, and it still reasoned at the level of individual intersections. Coordinating multiple agents across a corridor required a separate, manually designed orchestration layer. This limitation forced the final design transition.

3.3 Iteration Three: Cooperative Multi-Agent Reinforcement Learning

In Accra-Flow, each intersection becomes an autonomous reinforcement learning agent. The critical addition is cooperation: under Multi-Agent RL (MARL), agents share information about their state and are rewarded not for individual performance but for the health of the entire network. The system's objective is encoded in a single reward function:

$$R = w_1(CFI) + w_2(-D_queue) + w_3(-E_emissions) - \varphi(v_drift)$$

This equation simultaneously maximises passenger throughput across the network, minimises queue depth at all monitored junctions (preventing the Balloon Effect by penalising agents that export congestion), and minimises emissions from extended idling. The touting penalty from the TPA is retained.

The mechanism for solving the Balloon Effect is pressure propagation. Each intersection periodically broadcasts its congestion pressure to its immediate neighbours. If a downstream junction is already saturated, an upstream agent preemptively slows the release of vehicles into that segment. Coordination emerges from the learning process itself rather than from any static rule set.

4. Technology Stack

Tier 1: Perception

The system's eyes are purpose-built roadside computers, NVIDIA Jetson units, mounted at target intersections. The critical design choice is local processing: each unit analyses its camera feed on-device in real time, classifying vehicles by type, counting them, and tracking velocity without ever transmitting raw footage. Only anonymised metadata leaves the device. This is not merely a privacy preference. A system that streamed raw video would be slower, more expensive, and architecturally incompatible with Ghana's Data Protection Act 2012. Privacy compliance is built into the hardware layer because it was treated as an engineering constraint from the start, not a legal checkbox applied afterward. The units

operate offline during power outages without losing local signal control, a non-negotiable requirement in the West African urban environment.

Tier 2: Reasoning

This is where the MARL agents live, where pressure propagation between intersections occurs, and where the system forecasts traffic conditions 5 to 30 minutes ahead to anticipate bottlenecks before they form. Agents communicate through a lightweight messaging protocol hosted on an edge cloud, allowing the full network to function as a coordinated team rather than isolated nodes. The geospatial foundation is OpenStreetMap, chosen specifically because it is the only mapping platform that maintains stop-level data for informal TroTro networks. Proprietary providers do not cover paratransit at this resolution. Using them would mean building a system for a city that the maps do not actually represent.

A fundamental challenge in deploying machine learning in Accra is the near-absence of historical traffic data. Accra-Flow's response is a Synthetic-to-Real (S2R) pipeline: MARL agents are trained in high-fidelity simulations before any hardware is deployed. The environment combines SUMO (Simulation of Urban Mobility) for real road geometries with over 90% agreement against observed N4 corridor behaviour, DesRUTGe for realistic synthetic traffic generation, and NVIDIA Omniverse Digital Twins for risk-free testing of infrastructure changes before physical construction begins.

Tier 3: Interaction

The interaction tier solves the last-mile problem of the entire framework: getting decisions out of the system and into the hands of drivers who need to act on them. Two choices reflect the specific realities of Greater Accra's digital infrastructure.

USSD alerts require no internet connection, no data plan, and no smartphone. When the system identifies a demand hotspot or recommends an alternative route, the message reaches the driver's handset the same way an airtime balance check does, instantly and on any device.

The MoMo Fuel Rebate Token addresses a subtler barrier. For a driver whose fuel cost represents roughly 40% of gross revenue, an AI-recommended detour to an unfamiliar corridor is not just inconvenient: it is a financial risk. The token system transfers that risk from the driver to the framework. Follow the recommended route, demonstrate adherence, and receive a guaranteed fuel credit to your mobile money wallet. This single mechanism converts driver behaviour from an obstacle into an asset.

5. Projected Impact

Accra-Flow's projected outcomes are derived from the SUMO simulation environment calibrated against field measurements on the N4 corridor and from published evidence on comparable interventions in peer cities.

Metric	Baseline	Projected Post-Deployment	Basis
Average peak journey time, Madina-Tetteh Quarshie	67 minutes	48-54 minutes	SUMO simulation, N4 calibration
Annual economic loss recoverable	$301.5 million total	$45-72 million in Phase 1-2 scope	15-24% throughput improvement applied to time and fuel cost categories
PM2.5 contribution from idling (corridor)	Index 100	Index 78-85	Reduced idle time modelled against emissions coefficients [4]
TroTro driver fuel expenditure as share of revenue	~40%	~34-36%	MoMo rebate coverage plus reduced idling time

Table 4: Projected impact metrics. Projections assume 60% driver participation and do not account for network effects from full corridor coordination.

These projections are conservative. They assume partial adoption, specifically 60% driver participation in the USSD and MoMo systems, and do not account for network effects that emerge from corridor-level coordination beyond the two primary N4 chokepoints. Full city-wide deployment, which the Phase 4 roadmap targets via Circle interchange, would be expected to unlock proportionally larger gains. Independent validation against live traffic data is built into the Phase 2 and Phase 3 evaluation protocols.

6. Implementation Roadmap: Madina-Tetteh Quarshie Corridor

The Madina-Tetteh Quarshie corridor (the N4 road) was selected as the primary pilot testbed because field data identifies it as one of Greater Accra's most productivity-damaging routes, and it is currently undergoing active interchange expansion, a live, evolving environment that stress-tests the system's adaptive capabilities from day one.

Phase	Activity
Phase 1: Grounding	Real-time GPS data collected from instrumented TroTros on the N4. SUMO digital twin calibrated against field measurements. OpenStreetMap geometries validated to incorporate interchange reconfigurations not yet reflected in formal map providers.
Phase 2: Logic Deployment	NVIDIA Jetson units deployed at Oponglo junction and Madina Police Station junction. TPA activated as rule-based baseline. USSD gateway load-tested with recruited drivers. MoMo token integration validated end-to-end.
Phase 3: Stress Testing	Ghost Buses (virtual prioritised transit vehicles) injected into the SUMO simulation as counterfactual audit tools, isolating the causal effect of signalling interventions on journey times and emissions across 24-hour synthetic traffic cycles.
Phase 4: Expansion	37 Military Hospital Junction targeted for emergency-vehicle priority integration. Circle (Kwame Nkrumah Interchange) serves as the ultimate scalability test: success there validates the framework for city-wide deployment.

Table 5: Four-phase implementation roadmap for the N4 pilot corridor.

7. Comparative Evidence and Policy Framework

Lessons from Peer Cities

The Digital Matatus project in Nairobi mapped the routes and stops of that city's informal minibus network, a transit typology nearly identical to Accra's TroTro system, and released the data openly. Local entrepreneurs built routing applications without any coordination with the project team, creating a self-sustaining data ecosystem. Accra-Flow adopts the same open-data philosophy, using OpenStreetMap as the community-maintained foundation rather than proprietary APIs.

Research on Kigali's bus network found that performance-based driver compensation leads to higher passenger turnover but at the cost of faster, less safe driving [23, 24]. This finding directly shaped the MoMo incentive design: tokens reward adherence to route and safety-consistent speed profiles, not passenger counts. Driver behaviour is treated as an endogenous variable that the system shapes, not an exogenous constraint it works around.

Data Governance and Sovereignty

The Accra-Flow architecture was designed from the outset for compliance with Ghana's Data Protection Act 2012 (Act 843). All video processing occurs on the edge device; no raw footage is ever transmitted. The system's outputs are anonymised aggregates with no individual journey histories. This is not merely a legal checkbox. A system that collected personal mobility data would face significant community resistance; privacy compliance is an architectural constraint that enables adoption.

Every component uses open-source or open-data alternatives, ensuring that no single vendor's pricing decision or service discontinuation can compromise the system. Greater Accra's urban mobility data is treated as infrastructure data: its governance must remain under Ghanaian institutional oversight.

8. Conclusion

The Accra-Flow development journey followed a consistent pattern: each iteration revealed not a flaw in the solution, but an underestimation of the problem's scope. The CFI correctly identified people as the right unit of optimisation but could not act on that insight at the network level. The TPA translated the CFI into actionable signal logic but could not adapt to conditions outside its design envelope. The MARL system transcends these constraints by allowing the network to discover optimal coordination policies through experience, but only because the preceding iterations had correctly defined what optimal means in Greater Accra's specific context.

The framework's central claim is validated by the architecture itself: Accra's transit crisis is an informational failure masquerading as an infrastructure problem. The infrastructure was always there. What was missing was the intelligence to coordinate it.

The $301.5 million to be reclaimed annually is not merely a financial figure. It represents ten hours per week returned to students like Abena, perishable inventory preserved for traders like Auntie Efua, and fuel margins restored to drivers like Kojo. When urban mobility infrastructure fails, it fails people. When it succeeds, it returns to them the most finite resource they have: time.

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